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Rapid degradation of sulfur-containing pollutants in water using potassium oximate through nucleophilic transformation with kinetic, mechanistic and antimicrobial evaluation

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Sulfur-containing organic pollutants persist in industrial wastewater, particularly in petroleum refining effluents, where their chemical stability limits conventional treatment. Here, we report a rapid and efficient degradation strategy based on potassium 2,3-butanedione mono-oximate (KBDO), using 2-Chloroethyl ethyl sulfide (CEES) as a representative thioether model. Under mild aqueous conditions (room temperature, atmospheric pressure, aqueous medium, and pH 10), KBDO enables fast and selective transformation of CEES, achieving up to 97% removal in real petroleum wastewater. Kinetic analysis reveals pseudo-first-order behavior with a high-rate constant ($k = 0.495 \text{ min}^{-1}$) and a short half-life (1.4 min), indicating exceptionally rapid reaction dynamics. Product analysis by GC–MS identifies multiple transformation products consistent with an oximate-driven nucleophilic substitution pathway followed by fragmentation and cyclization. Notably, the system also exhibits antimicrobial activity across a range of microbial strains, demonstrating dual functionality. Unlike conventional oxidation-based treatments, this approach operates without catalysts or external energy input, offering a simple and environmentally benign alternative. These findings establish KBDO as a promising platform for fast, scalable, and multifunctional remediation of sulfur-containing pollutants in complex water systems.

Keywords Sulfur pollutants, CEES, Oximate, Degradation, Kinetics, Wastewater, Antimicrobial activity

Sulfur-containing organic compounds are widely encountered in industrial waste streams generated from petroleum refining, natural gas processing, and the manufacture of agrochemicals and specialty chemicals^{1–3}. Among these pollutants, thioether and sulfide derivatives represent a particularly challenging class due to their persistence, strong odor, and potential toxicity in aquatic environments^{4–6}. Petroleum refinery wastewater is known to contain a complex mixture of organic contaminants including sulfides, mercaptans, and heterocyclic sulfur compounds such as thiophenes, which often exhibit high resistance toward conventional biological treatment processes^{7–9}. Their chemical stability and limited biodegradability allow them to persist in environmental matrices, posing long-term risks to ecosystems and water resources^{10,11}. Several studies have reported the presence and environmental impact of sulfur-containing organic compounds in refinery wastewater. For example, Wake reported that petroleum refinery effluents contain a diverse range of sulfur-based pollutants that can significantly affect aquatic ecosystems if not properly treated¹². Similarly, Diya'uddeen et al. reviewed treatment technologies for petroleum refinery wastewater and highlighted that sulfur-containing compounds remain among the most difficult pollutants to remove using conventional treatment processes¹³. Advanced analytical investigations have also confirmed the presence of complex sulfur-containing organic

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species in refinery effluents using mass spectrometric techniques¹⁴. These findings underline the need for efficient chemical transformation strategies capable of degrading persistent sulfur-based contaminants in aqueous environments. Various chemical and catalytic approaches have been explored to degrade sulfur-containing organic pollutants in water¹⁵. Oxidative processes are among the most commonly investigated techniques. For instance, Zhang et al. examined the oxidation of odorous sulfide compounds in drinking water using oxidants such as NaClO, ClO₂, and KMnO₄, demonstrating effective degradation but requiring relatively strong oxidants and extended reaction times¹⁶. Photocatalytic degradation has also been reported as a promising approach. Wang et al. studied the photocatalytic oxidation of dimethyl sulfide using TiO₂-based catalysts under UV irradiation, achieving significant pollutant removal; however, the system relied on external energy input and catalyst activation¹⁷. More recently, Li et al. reported rapid detoxification of sulfur mustard simulants using advanced catalytic materials based on covalent organic frameworks supported with silver nanostructures, although such systems involve complex synthesis procedures and light-assisted activation¹⁸. In addition, catalytic oxidation of heterocyclic sulfur compounds such as thiophene using modified TiO₂ systems has been investigated, highlighting the potential of photocatalytic systems for sulfur pollutant degradation¹⁹. Despite these advances, many reported systems rely on strong oxidants, complex catalysts, or external irradiation sources, which may limit their practical application in large-scale water treatment processes. Moreover, most studies primarily focus on removal efficiency while providing limited mechanistic insight or kinetic evaluation under mild aqueous conditions (room temperature, atmospheric pressure, aqueous medium, and pH 10). Although oxime-based nucleophiles have been extensively investigated for the detoxification of sulfur mustard (HD) and its simulants in military decontamination and chemical defense applications, these studies have primarily focused on rapid decontamination of contaminated surfaces, protective materials, or emergency neutralization systems rather than environmental water treatment. For example, Yang et al. demonstrated the high nucleophilic reactivity of oxime-based compounds toward sulfur mustard simulants²⁰, while Wagner et al. reported efficient detoxification strategies employing reactive oxime formulations for chemical warfare agent decontamination²¹. Despite these important advances, the application of oxime chemistry as an environmentally oriented water-treatment technology, particularly for degradation kinetics, mechanistic investigation, antimicrobial assessment, and validation using real industrial wastewater, remains largely unexplored. Therefore, the present work extends oxime chemistry beyond conventional military decontamination by establishing an integrated environmental remediation strategy based on potassium 2,3-butanedione monooximate (KBDO). To investigate degradation mechanisms of reactive thioether pollutants under controlled laboratory conditions, surrogate compounds are often employed instead of highly toxic vesicants. In this context, 2-Chloroethyl ethyl sulfide (CEES) is widely used as a representative model compound because it preserves the characteristic thioether functionality and electrophilic reactivity of sulfur-containing pollutants while presenting significantly lower toxicity than sulfur mustard. Consequently, CEES has been extensively used to investigate degradation pathways and detoxification strategies relevant to sulfur-based contaminants. Oxime-derived reagents have attracted considerable attention due to their strong nucleophilic properties and their ability to react with electrophilic centers in sulfur-containing molecules, leading to rapid chemical transformation and detoxification^{22,23}. Nevertheless, the application of oxime-based systems for environmental remediation of thioether pollutants in aqueous media remains relatively underexplored, particularly in relation to quantitative kinetic analysis and their performance in real wastewater systems. In addition to chemical contamination, industrial wastewater streams often contain significant microbial populations that may contribute to secondary pollution or biofilm formation²⁴. Therefore, treatment approaches capable of addressing both chemical pollutants and microbial contamination are highly desirable. In this context, compounds exhibiting both pollutant degradation capability and antimicrobial activity may provide multifunctional solutions for water remediation. In the present study, the degradation of CEES using potassium 2,3-butanedione mono-oximate (KBDO) is investigated as an efficient nucleophilic reagent operating under mild aqueous conditions (room temperature, atmospheric pressure, aqueous medium, and pH 10). The potassium salt form enhances the solubility and reactivity of the oxime species in water, facilitating rapid interaction with the target pollutant. The degradation process was monitored using gas chromatography–mass spectrometry (GC–MS) to evaluate transformation products, degradation pathways, and reaction kinetics. To the best of our knowledge, a detailed kinetic study of CEES degradation using KBDO in aqueous systems has not been previously reported. Furthermore, the antimicrobial activity of the oximate system was evaluated against a panel of microbial strains in order to explore its potential dual functionality in water treatment applications. Finally, the applicability of the oximate reagent was investigated using a real petroleum refinery wastewater sample, enabling assessment of its performance in a complex environmental matrix. The results demonstrate rapid CEES degradation, significant pollutant removal efficiency, and notable antimicrobial activity, highlighting the potential of KBDO as a promising reagent for the simultaneous remediation of sulfur-containing pollutants and microbial contaminants in contaminated water systems. This work introduces, for the first time, a nucleophile-driven degradation platform combining rapid kinetics, mechanistic clarity, and antimicrobial functionality in real wastewater systems.

Materials and methods

Materials

All chemicals and solvents used in this study were of analytical grade and were used as received without further purification. 2,3-butanedione mono-oxime (≥98%) was obtained from Sigma-Aldrich. Potassium hydroxide (KOH) pellets (≥99%) were purchased from Merck. Ethanol (absolute, ≥99.8%) and diethyl ether (analytical grade) were supplied by Fisher Scientific. The sulfur-containing model compound 2-Chloroethyl ethyl sulfide (≥98%) was obtained from Sigma-Aldrich and used as received. Deionized water was used throughout all experiments for solution preparation.

Preparation of potassium salt of 2,3-butanedione mono-oxime

Figure (1) demonstrates the preparation steps of KBDO. A 250 mL alcoholic standard solution of 2,3-butanedione mono-oxime (solution 1) and a separate alcoholic solution of potassium hydroxide (solution 2) were prepared. Solution 1 was added slowly to Solution 2 at 25 °C over a period of 2 h with continuous stirring. Then the reaction mixture was heated indirectly to 80 °C and distilled under reduced atmospheric pressure with care to avoid thermal decomposition of the product. Finally, the solid residue obtained after distillation was washed with diethyl ether, followed by drying in a vacuum oven for 24 h. The yield of the final product was calculated and it was about 90.5%²⁵.

Fourier transform infrared spectroscopy (FT/IR)

FT/IR spectra of both the free oxime and the potassium salt were recorded using FT/IR (Model: Nicolet iS10, Thermoscientific, USA) equipped with an attenuated total reflectance (ATR) accessory. The spectra were collected in the range of 4000–400 cm⁻¹ at a resolution of 4 cm⁻¹. All samples were analyzed without further preparation.

Nuclear magnetic resonance (NMR) spectroscopy

¹H NMR and ¹³C NMR spectra were obtained for the free oxime and the potassium salt using an NMR spectrometer (Model: Varian 400 MHz, USA). Deuterated solvents D₂O and DMSO-d₆ were used, and chemical shifts (δ) were referenced to the residual solvent peak. Measurements were performed at room temperature under standard acquisition parameters²⁶.

X-ray diffraction (XRD) measurements

Powder XRD patterns were recorded using an X-ray diffractometer (Model: Dynamic 500, Anton paar, Switzerland) with Cu Kα radiation (λ = 1.5406 Å). Data were collected over a 2θ range of (20–90) ° with a step size of 0.02° and a scan rate of 0.5 s/step.

X-ray photoelectron spectroscopy (XPS)

Surface chemical composition and elemental states of the synthesized compounds were investigated using X-ray photoelectron spectroscopy (Model: K-Alpha+, Thermo Scientific, UK). The measurements were carried out using an XPS spectrometer equipped with a monochromatic Al Kα radiation source (hν = 1486.6 eV). Survey spectra were collected over a wide binding energy range to identify the elements present on the surface of the samples. The spectra were recorded with a pass energy of 100 eV and a step size of 1.0 eV, while the analysis area was approximately 350 μm. Charge correction was performed by referencing the C1s peak to 284.8 eV. The obtained spectra were processed and fitted using appropriate background subtraction and peak deconvolution procedures in order to determine the chemical states of the detected elements. Elemental compositions were estimated from the integrated peak areas using sensitivity factors provided by the instrument software²⁷.

Gas chromatography–mass spectrometry (GC–MS) analysis

The degradation behavior of CEES in the presence of KBDO was investigated under different experimental conditions. The experiments were performed in sealed glass reactors containing aqueous reaction mixtures under constant magnetic stirring at room temperature (25 ± 2 °C). To evaluate the influence of the KBDO concentrations, a series of experiments were conducted using different KBDO concentrations of 0, 5, 10, 15, and 20% (v/v) while maintaining a fixed dosage of the CEES. The reaction mixtures were allowed to proceed under identical conditions, and aliquots were withdrawn at predetermined time intervals to monitor the degradation process. The effect of solution pH on the degradation efficiency was also investigated. The pH of the reaction media was adjusted to the desired values (3, 7 and 10) using dilute HCl or NaOH solutions and measured using a calibrated digital pH meter. The reaction mixtures were maintained at the selected pH throughout the experiments in order to examine the influence of acidity and alkalinity on the degradation kinetics. Additionally, the effect of contact time was evaluated by sampling the reaction mixture at different time intervals during the degradation process. The collected samples were immediately subjected to liquid–liquid extraction prior to GC–MS analysis to quantify the residual CEES and identify the degradation products. For quantitative analysis, the reaction mixtures were extracted using n-hexane as the organic solvent. Typically, 10 mL of the reaction solution was transferred into a separatory tube and mixed with 5 mL of n-hexane. The mixture was vigorously shaken for approximately 3 min to ensure efficient partitioning of CEES and its degradation products into the organic phase. After phase separation, the upper organic layer was carefully collected and dried over anhydrous sodium sulfate to remove residual moisture. The extract was then filtered and transferred into GC vials for chromatographic analysis. The extracted samples were analyzed using a gas chromatography–mass spectrometry system consisting of an Agilent 7890B GC System coupled with an Agilent 5977 A MSD. Chromatographic separation was achieved using an HP-5MS capillary column (30 m × 0.25 mm internal diameter × 0.25 μm film thickness). Helium was used as the carrier gas at a constant flow rate of 1.0 mL min⁻¹. The injection volume was 1 μL in split mode (split ratio 10:1), and the injector temperature was maintained at 250 °C. The oven temperature program was set as follows: initial temperature 40 °C (held for 2 min), ramp at 10 °C min⁻¹ to 200 °C, then increased to 280 °C (held for 5 min). The mass spectrometer was operated in electron ionization (EI) mode at 70 eV with a scan range of m/z 40–400. The ion source and quadrupole temperatures were maintained at 230 °C and 150 °C, respectively. The degradation of CEES was evaluated by monitoring the decrease in the characteristic chromatographic peak corresponding to CEES. The resulting chromatograms were also used to identify intermediate and final degradation products based on their mass spectral fragmentation patterns²⁸.

Kinetic study

To evaluate the degradation behaviour of CEES in the presence of KBDO, the reaction kinetics were investigated by monitoring the temporal variation of CEES concentration using GC–MS analysis under the optimized experimental conditions. The concentration of CEES at different reaction times was determined from the corresponding chromatographic peak areas. The degradation efficiency was calculated based on the decrease in CEES concentration according to Eq. (1):

$$\text{Removal(\%)} = \frac{A_0 - A_t}{A_0} \times 100 \quad (1)$$

where A_0 represents the initial concentration of CEES and A_t is the concentration at reaction time t . The kinetic behaviour of CEES degradation was analysed by fitting the experimental data to a pseudo-first-order kinetic model, which is commonly applied for degradation reactions occurring in the presence of an excess amount of reactive species. The pseudo-first-order rate equation can be expressed as shown in Eq. (2):

$$\frac{dC}{dt} = -kC \quad (2)$$

Integration of the above equation gives the linearized form as shown in Eq. (3):

$$\ln\left(\frac{A_0}{A_t}\right) = kt \quad (3)$$

or equivalently Eq. (4):

$$\ln(A_t) = \ln(A_0) - kt \quad (4)$$

where k is the apparent pseudo-first-order rate constant (min^{-1}) and t is the reaction time. The rate constant was determined from the slope of the linear plot of $\ln(A_0/A_t)$ versus time. The half-life ($T_{0.5}$) of CEES degradation, defined as the time required for the pollutant concentration to decrease to half of its initial value, was calculated using Eq. (5):

$$T_{0.5} = \frac{\ln 2}{k} \quad (5)$$

where k is the pseudo-first-order rate constant obtained from the kinetic plot. The good linear correlation between $\ln(A_0/A_t)$ and reaction time confirms that the degradation of CEES in the presence of KBDO follows pseudo-first-order kinetics²⁹. The calculated kinetic parameters, including the rate constant and half-life values under different experimental conditions, are summarized in the corresponding tables and discussed in the Results section.

Antimicrobial activity assay

The antimicrobial activity of KBDO was evaluated against a panel of marine pathogenic microorganisms including Gram-positive bacteria, Gram-negative bacteria, and yeast. The tested strains included *Staphylococcus aureus*, *Escherichia coli*, *Pseudomonas aeruginosa*, *Bacillus subtilis*, *Enterococcus faecalis*, *Klebsiella pneumoniae*, *Vibrio cholerae*, *Candida albicans*, *Pseudomonas fluorescens*, *Streptococcus agalactiae*, and *Aeromonas hydrophila*. Each strain was cultured in nutrient broth and incubated until reaching the exponential growth phase. The inoculum density was standardized to approximately 10^5 – 10^6 CFU mL^{-1} to ensure reproducibility³⁰. The antimicrobial activity was assessed using a broth microdilution method, where KBDO solutions were prepared in a concentration range of 0.03–1.0 mg mL^{-1} ³¹. Following incubation, microbial growth was quantitatively determined by measuring the optical density at 600 nm (OD_{600}) using a UV–Vis spectrophotometer. Negative controls (untreated cultures) and positive controls (standard antimicrobial agent) were included to validate the assay. The minimum inhibitory concentration (MIC) was defined as the lowest concentration resulting in a significant reduction in microbial growth³². The inhibition efficiency was calculated according to Eq. (6).

$$\text{Inhibition(\%)} = [(\text{OD}_{\text{control}} - \text{OD}_{\text{sample}})/\text{OD}_{\text{control}}] \times 100 \quad (6)$$

Treatment of industrial wastewater sample

A real industrial wastewater sample was collected from a petroleum processing facility in order to evaluate the applicability of KBDO under realistic environmental conditions. The collected sample was stored in clean brown glass bottles at 4 °C prior to analysis. The organic constituents present in the wastewater were analyzed by GC–MS following the same extraction and analytical procedures described previously for the laboratory samples. Briefly, the wastewater sample was subjected to liquid–liquid extraction using n-hexane, and the organic extract was analyzed under identical GC–MS operating conditions to allow direct comparison with the experiments performed using the model compound. For treatment experiments, a measured amount of KBDO was added to the wastewater sample (50:1) and the mixture was stirred at room temperature for a predetermined contact time (5 min) and the collected petroleum refinery wastewater exhibited an initial pH of 7.3. Prior to treatment, the pH was adjusted to 10.0 ± 0.1 using 0.1 M NaOH in order to match the optimized reaction conditions established during the laboratory-scale degradation experiments. All subsequent degradation experiments were therefore conducted under identical alkaline conditions to ensure direct comparison between synthetic and real

wastewater systems. After treatment, the sample was extracted again with n-hexane and analyzed by GC–MS using the same analytical conditions described earlier. Microbial populations in raw and treated wastewater samples were evaluated using standard microbiological culture techniques. Nutrient agar (NA) was used for total bacterial count, while selective media including MacConkey, EMB, SS agar, TCBS, Cetrimide agar, and Mannitol salt agar were used for identification of specific microbial groups. Samples were incubated at 30 °C for 2–7 days to allow colony development. Microbial presence and reduction were assessed by comparing growth patterns before and after treatment^{33–35} Fig. 1.

Results and discussion

FT/IR analysis

Figure 2 illustrates the FT/IR spectra of both the free oxime and its potassium salt. For the free 2,3-butanedione mono-oxime, the FT/IR spectrum displayed a broad absorption band around 3200–3300 cm^{−1}, corresponding to the O–H stretching vibration of the oxime group (–NOH)³⁶. A characteristic absorption of the C=N–OH functionality was also observed near 1640–1660 cm^{−1}, along with bands at ~2920 and ~2850 cm^{−1} attributed to aliphatic C–H stretching³⁷. Additional absorptions were found at ~1450–1380 cm^{−1} (C–H bending) and at ~1050–1100 cm^{−1} (C–N stretching)³⁸. In contrast, the FT/IR spectrum of the potassium salt showed the disappearance of the O–H stretching band in the 3200–3300 cm^{−1} region, consistent with deprotonation of the oxime group. The C=NO[−] stretching vibration was shifted and appeared near 1580–1600 cm^{−1}, confirming salt formation. The aliphatic C–H stretching and bending vibrations remained unchanged, appearing at ~2920, 2850, and 1450–1380 cm^{−1}, respectively. These spectral differences, namely the loss of the –NOH stretching band and the shift of the C=NO resonance, provide further evidence for the successful conversion of the free oxime to its potassium salt. Table 1 illustrates the characteristic FT/IR bands of free monoxime and potassium salt oxime.

XRD analysis

The XRD diffraction patterns of the free 2,3-butanedione mono-oxime and its potassium salt are presented in figure 3. The free oxime illustrated in figure 3.a exhibited a semi-amorphous profile, characterized by a broad background with weak reflections³⁹. In contrast, the potassium salt illustrated in figure 3.b displayed sharper and more intense peaks, confirming the enhancement of crystallinity upon salt formation. Distinct diffraction peaks were observed for the salt at 2θ ≈ 9.8°, 16.3°, 41.3°, 48.0°, 50.1°, and 79.5°, which were absent or only weakly expressed in the free oxime. The emergence of these new crystalline reflections provides clear evidence of structural reorganization resulting from the deprotonation of the oxime and subsequent coordination with potassium. These findings are consistent with the FT/IR analyses, where the disappearance of the –NOH proton and the chemical shift of the oxime carbon resonance confirmed the conversion of the free oxime to its potassium salt. Collectively, the XRD data demonstrate that salt formation not only alters the molecular environment but also enhances the degree of crystallinity, supporting the successful synthesis of the KBDO.

NMR spectroscopic analysis

The structures of both the free 2,3-butanedione mono-oxime and its potassium salt were confirmed by ¹H and ¹³C nuclear magnetic resonance spectroscopy as demonstrated in figure 4. For the free oxime, the ¹H NMR spectrum shown in figure 4.a displayed a distinct downfield resonance at δ ≈ 13.9 ppm, corresponding to the –NOH proton, in addition to the expected aliphatic proton signals between δ (1.0–3.0) ppm. In the ¹³C NMR spectrum shown in figure 4.c, characteristic resonances were observed at δ ≈ 39–40 ppm (methyl and methylene carbons) and δ ≈ 155 ppm, attributable to the oxime carbon (C=NOH). In contrast, the ¹H NMR spectrum of the potassium salt (Fig. 4.b) revealed the complete disappearance of the –NOH resonance at δ ≈ 13.9 ppm, consistent with deprotonation and formation of the KBDO. The ¹³C NMR spectrum (Fig. 4.d) showed a similar pattern of aliphatic signals, but the oxime carbon resonance exhibited a noticeable chemical shift compared to the free oxime (from δ ≈ 155 ppm to δ ≈ 160 ppm), confirming salt formation. These spectral changes, namely the loss of the –NOH proton signal and the downfield shift of the oxime carbon resonance, provide strong evidence for the successful conversion of the free oxime to its potassium salt⁴⁰.

X-ray photoelectron spectroscopy (XPS)

Figure 5 demonstrates the X-ray photoelectron spectroscopy for both the free oxime and KBDO to investigate the surface chemical composition and electronic environments of the synthesized compounds. The survey spectrum of the free 2,3-butanedione mono-oxime (Fig. 5.a) reveals three dominant elements, namely carbon, nitrogen, and oxygen, which correspond to the expected molecular structure of the oxime compound. Quantitative analysis indicates that carbon represents the major surface component with an atomic percentage of approximately 67.93%, followed by oxygen (18.08%) and nitrogen (13.01%), as summarized in Table 2. The C1s peak observed at 284.86 eV corresponds mainly to aliphatic carbon atoms (C–C/C–H), whereas the N1s peak located at 399.98 eV is characteristic of the oxime nitrogen associated with the C=N–OH functional group⁴¹. The O1s signal centered at 531.01 eV originates from oxygen atoms within the hydroxylated oxime environment. Upon conversion to the potassium salt, the survey spectrum (Fig. 5.b) exhibits an additional peak corresponding to the K2p core level, confirming the successful incorporation of potassium into the molecular framework. The atomic concentration of potassium was estimated to be approximately 12.54%, accompanied by noticeable variations in the relative proportions of carbon, nitrogen, and oxygen. In particular, the nitrogen atomic percentage decreases from 13.01% to 7.64%, while oxygen increases to approximately 25.65%, as listed in Table 2. These changes reflect the deprotonation of the hydroxyl group and the subsequent coordination of potassium ions with oxygen atoms, leading to the formation of KBDO. High-resolution XPS spectra were further analyzed to elucidate the chemical environments of oxygen and potassium atoms. The O1s spectrum of the free oxime (Fig. 5.c) exhibits a dominant peak centered at 532.53 eV, which can be attributed to oxygen atoms in the oxime hydroxyl group

PREPARATION OF POTASSIUM SALT OF 2,3-BUTANEDIONE MONO-XIME

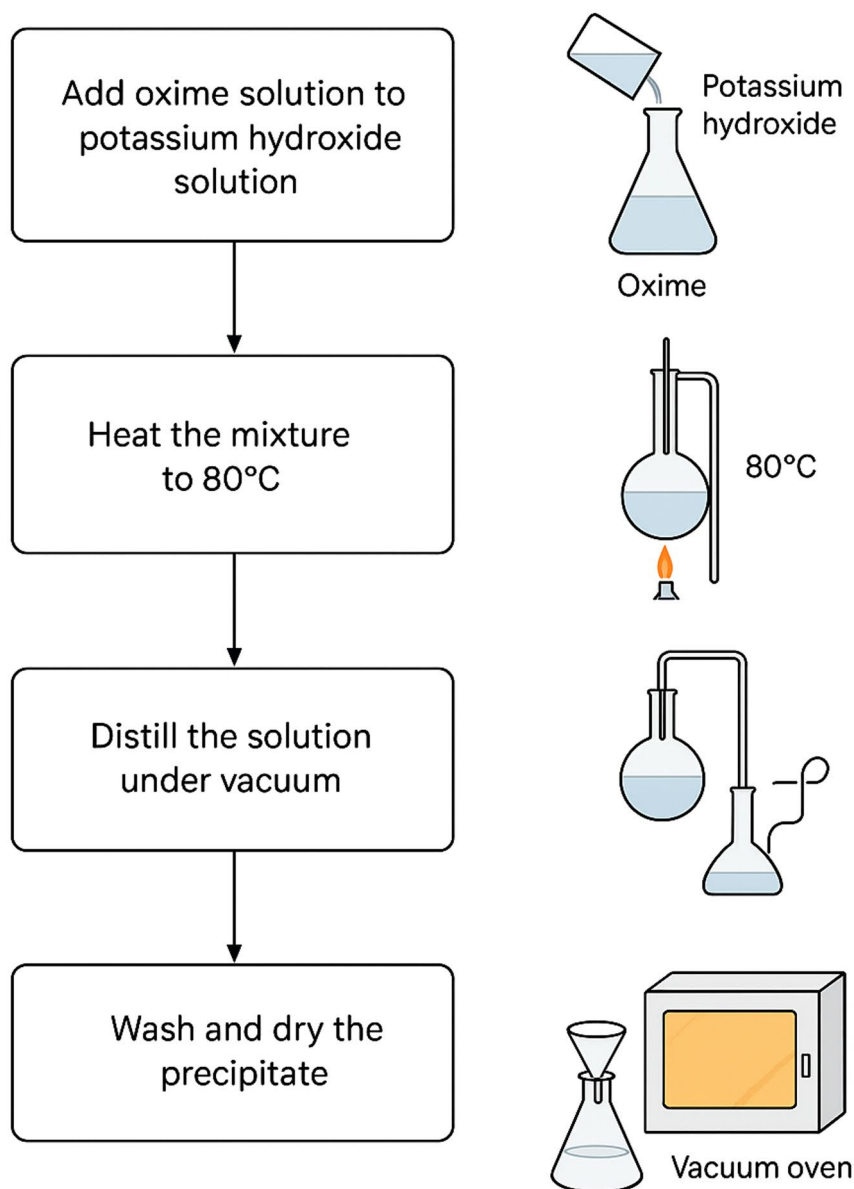


Fig. 1. Schematic illustration of the synthesis procedure of potassium 2,3-butanedione mono-oximate (KBDO).

(–NOH). Deconvolution of the spectrum reveals contributions corresponding to N–O bonding and hydroxyl oxygen within the oxime functionality. The binding energies and assignments of these peaks are summarized in Table 3. In contrast, the O1s spectrum of the KBDO (Fig. 5.d) displays a noticeable shift toward lower binding energies and can be resolved into two main components. The first component, located at 531.01 eV, is attributed to oxygen atoms coordinated with potassium (O–K), which arise from the deprotonation of the oxime hydroxyl group. The second component, appearing at 532.53 eV, corresponds to oxygen within the N–O environment of

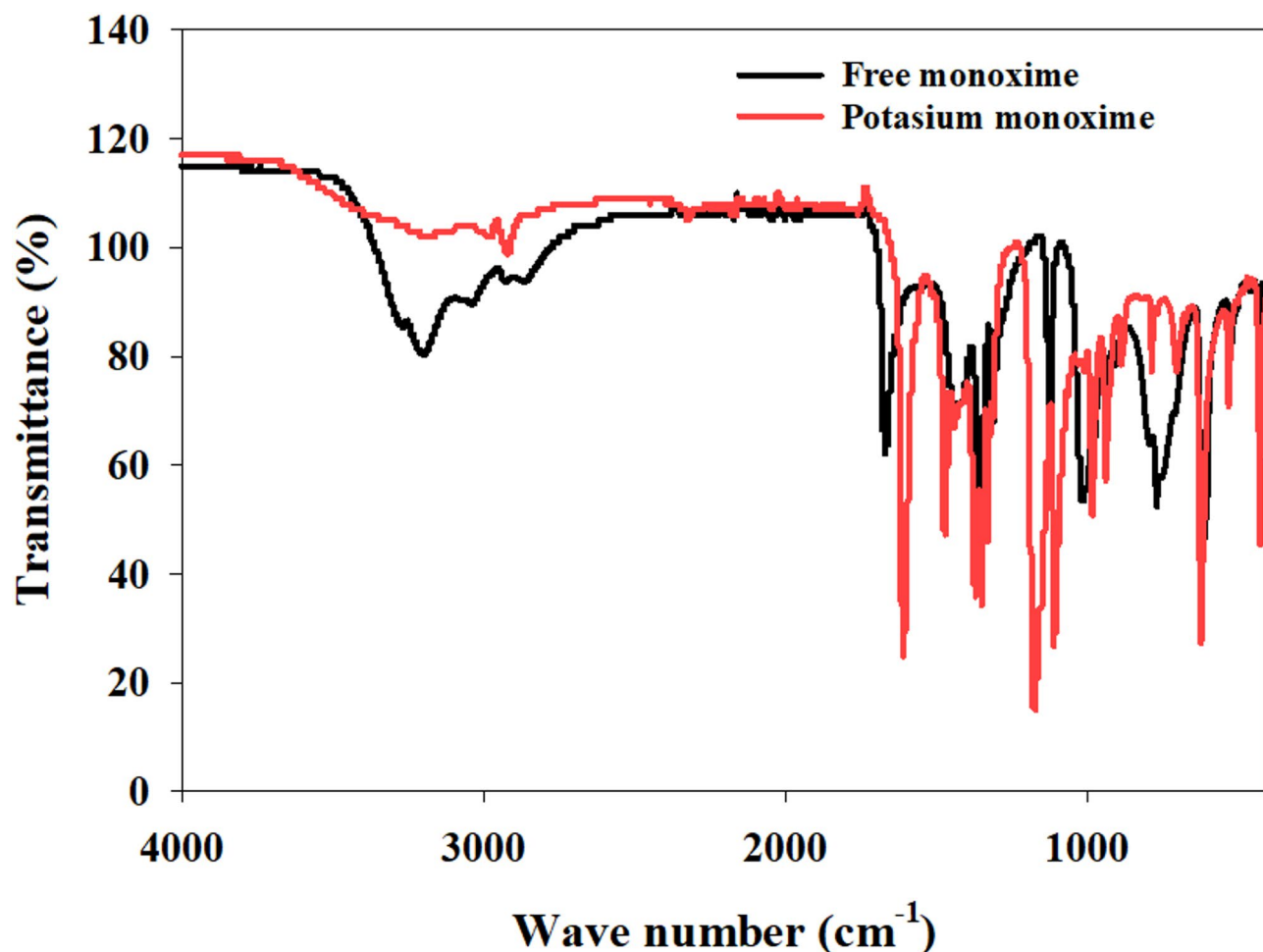


Fig. 2. FT/IR spectra of (a) free 2,3-butanedione mono-oxime and (b) its potassium salt (KBDO).

Functional groups	Free oxime (cm ⁻¹)	Potassium salt (cm ⁻¹)	Note
O–H stretching (–NOH)	3200–3300 (broad)	–	Disappearance indicates deprotonation
C=NOH stretching	1640–1660	–	Oxime functional group
C=NO ⁺ stretching	–	1580–1600	Shift confirms salt formation
C–H stretching (aliphatic, CH ₃ /CH ₂)	2920, 2850	2920, 2850	–
C–H bending	1450–1380	1450–1380	–
C–N stretching	1050–1100	1050–1100	–

Table 1. The characteristic FT/IR bands of the free monoxime and KBDO.

the oxime backbone. These binding energy values and their corresponding chemical assignments are listed in Table 3. Further confirmation is provided by the high-resolution K2p spectrum (Fig. 5.e), which exhibits the characteristic spin–orbit doublet of potassium with peaks located at 293.13 eV (K2p_{3/2}) and 296.04 eV (K2p_{1/2}). The energy separation of approximately 3 eV agrees well with previously reported values for potassium species coordinated with oxygen atoms⁴². The peak assignments and chemical states derived from the K2p analysis are also summarized in Table 3. The symmetric peak shapes and the absence of additional potassium-related features indicate that potassium exists predominantly in a single chemical environment within the oximate structure. Collectively, these results confirm the successful transformation of the neutral oxime into its potassium salt through deprotonation of the hydroxyl group and formation of ionic K–O interactions.

Degradation experiments and GC–MS analysis

Figure 6 illustrates the GC–MS analysis and the degradation experiments of CEES. The influence of KBDO concentrations on the degradation efficiency was systematically investigated under identical experimental conditions while maintaining a constant volume ratio between CEES and KBDO (1:50). The experiments were

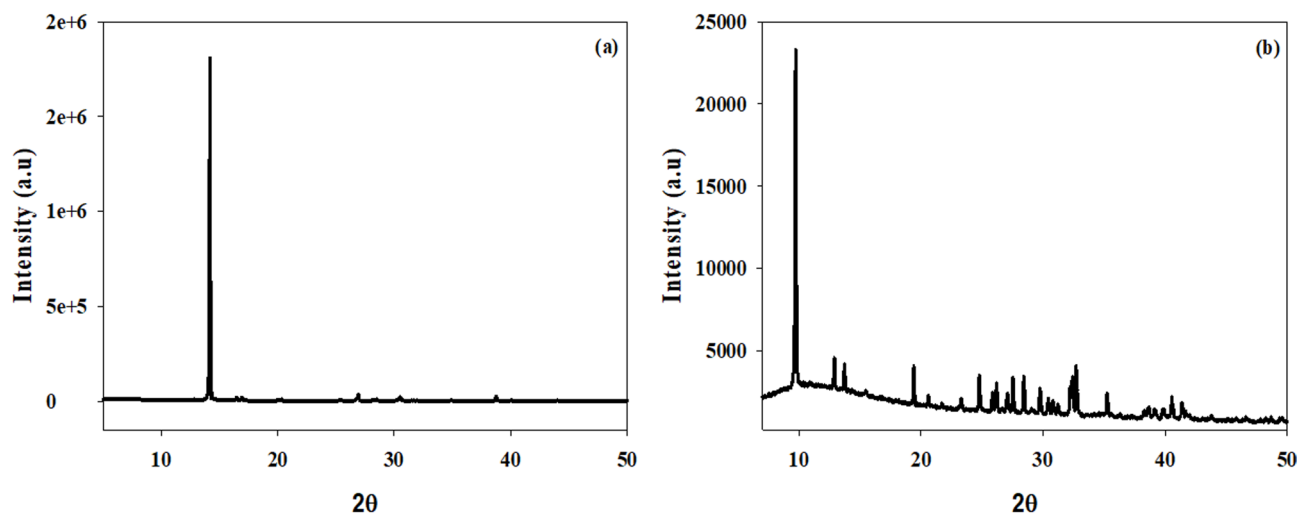


Fig. 3. XRD patterns of (a) free oxime and (b) potassium oximate (KBDO).

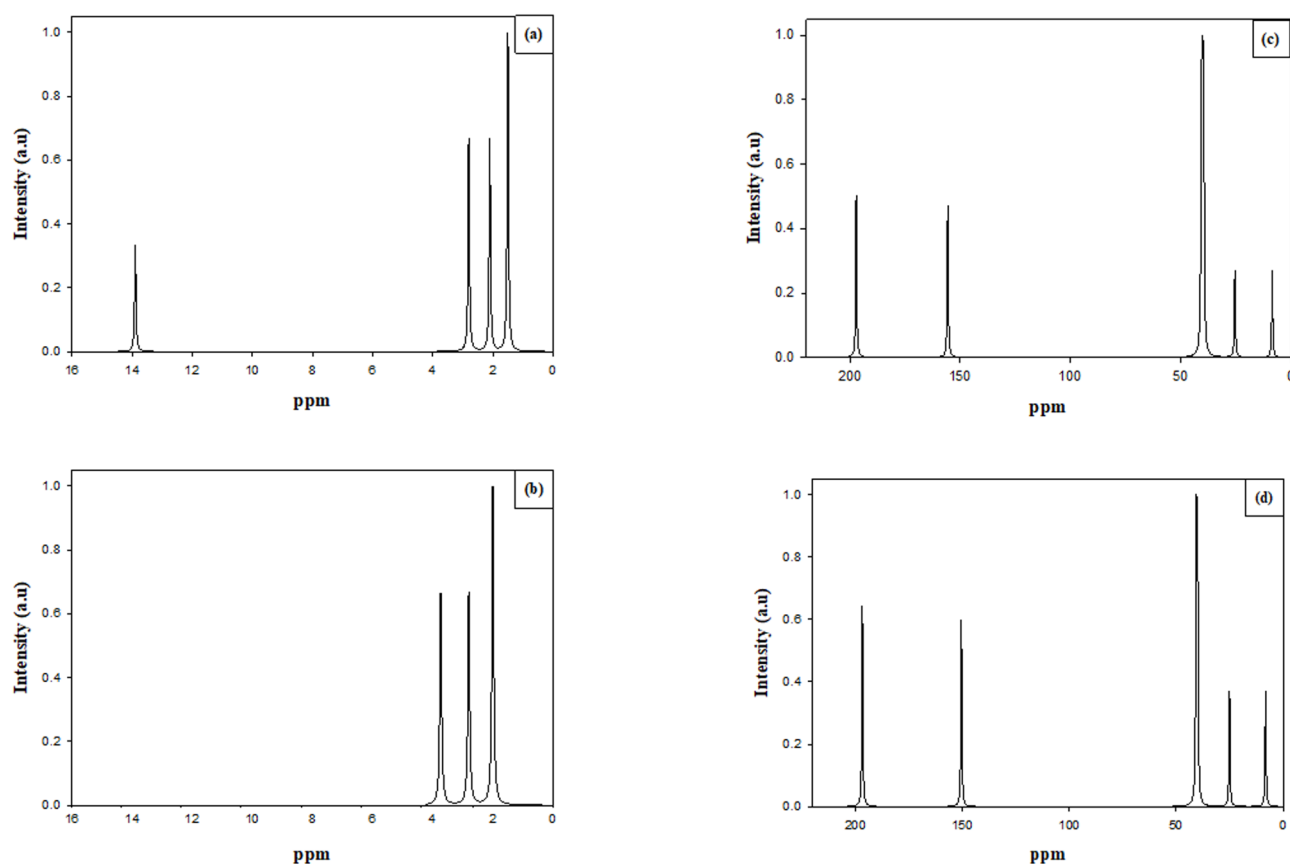


Fig. 4. NMR spectra of free oxime and potassium oximate (KBDO): (a) ¹H NMR of free oxime, (b) ¹H NMR of potassium salt, (c) ¹³C NMR of free oxime, and (d) ¹³C NMR of potassium salt.

performed using different initial KBDO concentrations of 0, 5, 10, 15, and 20%, and the degradation progress was monitored using GC–MS analysis. The corresponding degradation profiles are illustrated in figure 6. As shown in figure 6.a, the degradation efficiency increased gradually with increasing KBDO concentrations. At higher concentration, particularly 20%, the degradation proceeded rapidly, and a significant reduction in CEES concentration was observed within a relatively within 5 min. This behaviour can be attributed to the higher availability of active reactive species relative to the pollutant molecules, allowing more efficient interaction between CEES molecules and the oximate reagent⁴³. In contrast, when the KBDO concentration decreased to

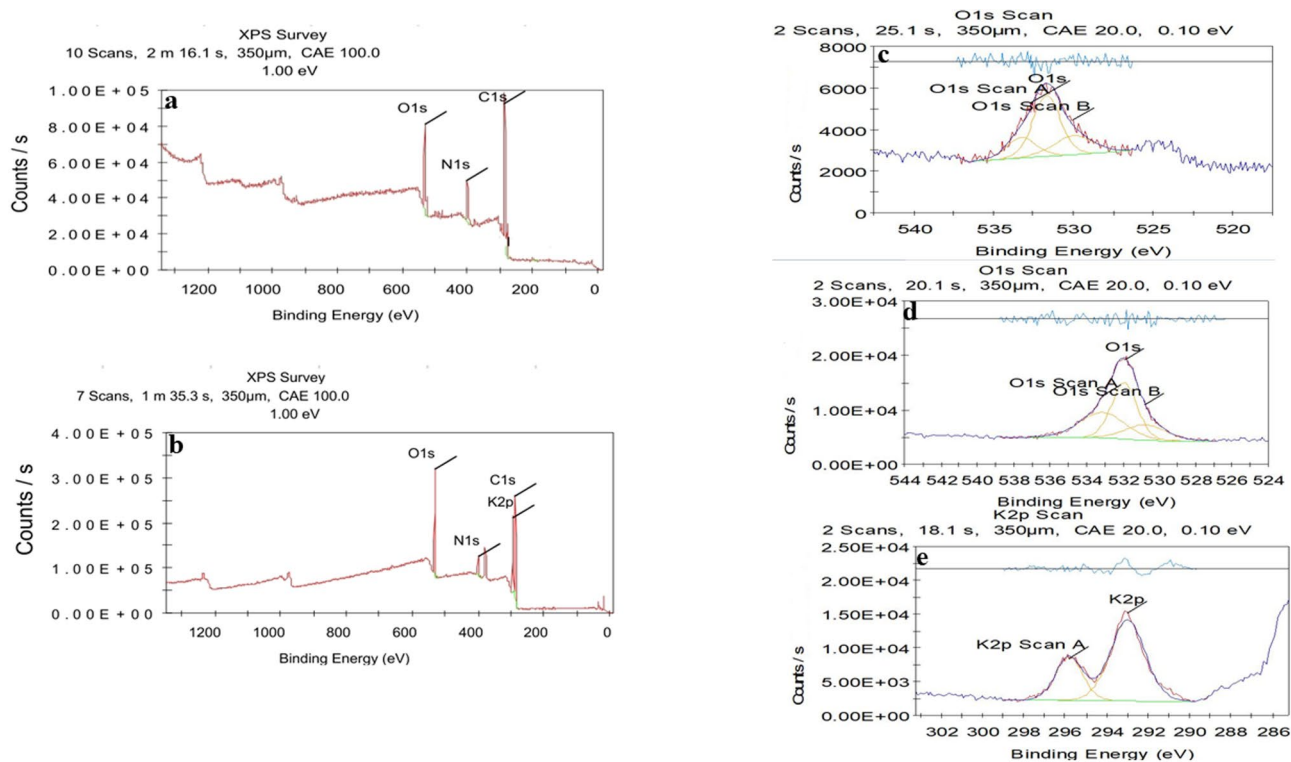


Fig. 5. XPS analysis of (a) survey spectrum of free oxime, (b) survey spectrum of potassium oximate (KBDO), (c) high-resolution O1s spectrum of free oxime, (d) high-resolution O1s spectrum of potassium oximate, and (e) high-resolution K2p spectrum.

Sample	C (at%)	O (at%)	N (at%)	K (at%)
Free oxime	67.93	18.08	13.01	—
KBDO	54.17	25.65	7.64	12.54

Table 2. Surface elemental composition.

Core level	Binding energy (eV)	Assignment	Chemical state
C1s	284.86	Aliphatic carbon	C–C/C–H
N1s	399.98	Oxime nitrogen	C=N
O1s	532.53	Oxime oxygen	N–O
O1s	531.01	Oxygen coordinated with potassium	O–K
K2p _{3/2}	293.13	Potassium ion	K ⁺
K2p _{1/2}	296.04	Spin–orbit component	K ⁺

Table 3. XPS peak assignment.

pH	Estimated CEES Peak Area (%)	Degradation (%)
3	~70	~30
7	~55	~45
10	~5	~95

Table 4. Estimated relative CEES peak areas at different pH values.

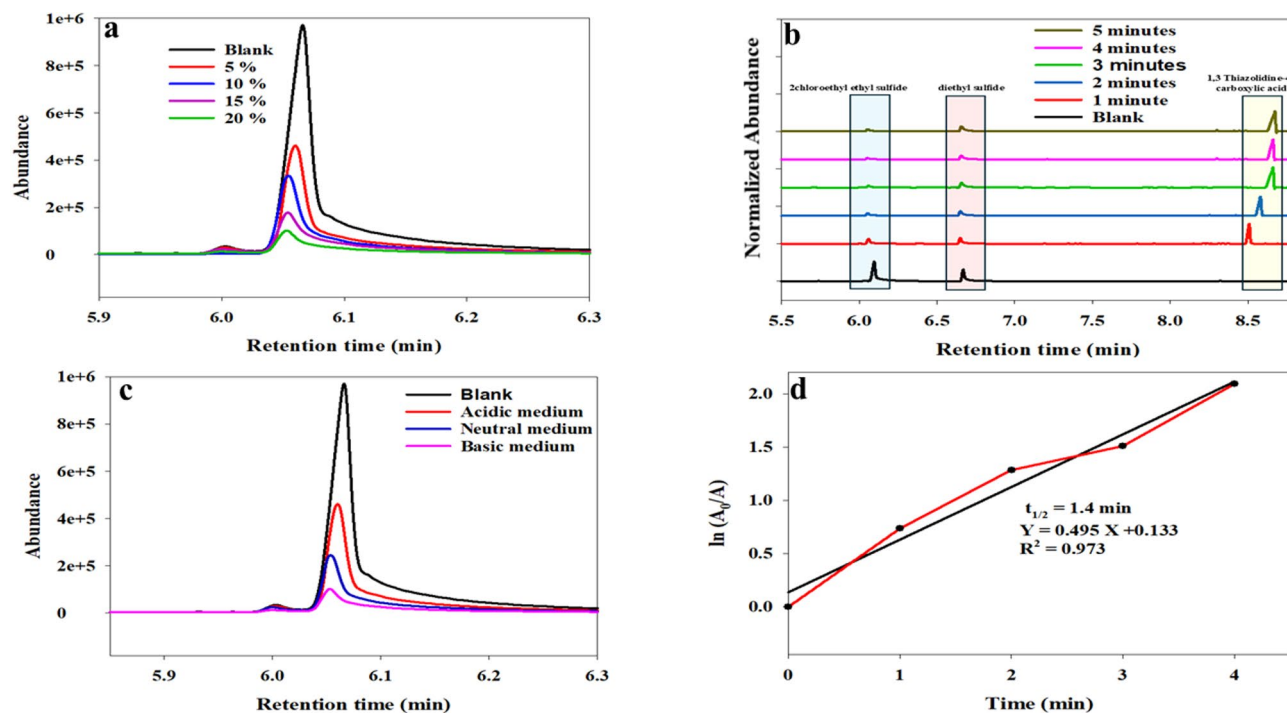


Fig. 6. GC-MS analysis of CEES degradation in the presence of potassium oximate (KBDO): (a) effect of initial KBDO concentration, (b) effect of contact time showing the progressive disappearance of CEES and formation of degradation products, (c) effect of pH on degradation efficiency, and (d) pseudo-first-order kinetic plot for CEES degradation.

15%, 10%, and 5%, the degradation rate became progressively slower. The lower KBDO concentration results in decreased availability of reactive species of the oximate reagent, thereby reducing the overall degradation efficiency. Additionally, the accumulation of intermediate products generated during the degradation process may partially inhibit the reaction progress by occupying active reaction sites or altering the local chemical environment. Furthermore, the results indicate that the degradation process follows a concentration-dependent behaviour, where the ratio between the oxidizing reagent and the pollutant plays a crucial role in determining the overall degradation performance. At lower KBDO concentrations, the fixed amount of KBDO becomes relatively insufficient to completely react with all pollutant molecules within the same reaction time, leading to a lower removal percentage⁴⁴. The present study optimized the degradation process based on KBDO concentration rather than stoichiometric or molar ratio optimization. Under the investigated conditions, a KBDO concentration of 20% provided complete CEES degradation within the shortest reaction time and was therefore selected for all subsequent experiments. Since higher KBDO concentrations were not investigated, it cannot be concluded whether degradation efficiency would further improve beyond 20%. Likewise, determination of the minimum stoichiometric requirement between KBDO and CEES was beyond the scope of the present study and warrants systematic investigation in future work. Overall, these findings demonstrate that higher KBDO concentrations favor more efficient degradation kinetics, highlighting the importance of optimizing the KBDO concentration for achieving maximum removal efficiency in practical remediation applications. The effect of contact time on the degradation of CEES was investigated by monitoring the reaction progress using GC-MS analysis, and the resulting chromatograms are presented in figure 6.b. The chromatographic profiles clearly demonstrate the gradual transformation of the parent CEES compound into several degradation products as the reaction time increases. As illustrated in figure 6.b the chromatogram of the blank sample is dominated by the characteristic peak of CEES at a retention time of approximately 6.0 min. Upon introducing the KBDO reagent with concentration of 20%, the intensity of the CEES peak progressively decreases with increasing reaction time from 1 to 5 min, indicating the continuous consumption of the parent compound during the degradation process. Simultaneously, new peaks corresponding to degradation products begin to appear and increase in intensity. Among the identified products are Diethyl sulfide, which appears at a retention time of approximately 6.6 min, and 1,3-Thiazolidine-4-carboxylic acid detected at around 8.6 min. The progressive increase in the abundance of these products with reaction time confirms the ongoing transformation of CEES into new chemical species. The formation of diethyl sulfide suggests cleavage or rearrangement within the thioether structure of CEES, while the appearance of 1,3-thiazolidine-4-carboxylic acid indicates that nitrogen-containing species from the oxime reagent participate in the reaction pathway, leading to the formation of heterocyclic sulfur-containing products. The increasing intensity of these peaks with time provides clear evidence for the effective chemical degradation of CEES in the presence of KBDO⁴⁵. The influence of solution pH on the degradation efficiency of CEES was investigated at pH values of 3, 7, and 10, and the corresponding GC-MS chromatograms are

presented in figure 6.c. The degradation performance was quantitatively evaluated by comparing the integrated peak area of CEES under each pH condition relative to the initial chromatogram. As shown in figure 6.c, the chromatographic peak corresponding to CEES (retention time ≈ 6.0 min) exhibits a strong dependence on the pH of the reaction medium. Under acidic conditions (pH = 3), the chromatogram is dominated by the CEES peak, indicating minimal degradation of the parent compound. Only a partial decrease in peak area is observed, suggesting that the majority of the pollutant remains unchanged. When the reaction medium is adjusted to neutral conditions (pH = 7), a better reduction in the CEES peak area is observed. This decrease indicates that a considerable fraction of the parent compound undergoes chemical transformation. A significantly stronger degradation effect is observed under alkaline conditions (pH = 10). In this case, the peak area corresponding to CEES decreases dramatically compared with the acidic and neutral conditions. The substantial reduction in peak intensity indicates that a large portion of the initial CEES has been converted into secondary products. The quantitative comparison of CEES peak areas estimated from the chromatographic profiles is summarized in Table 4. The data clearly demonstrate that the degradation efficiency increases progressively with increasing pH, reaching its maximum under alkaline conditions.

The degradation kinetics of CEES in the presence of KBDO were investigated in order to better understand the reaction rate and the underlying degradation behaviour. To the best of our knowledge, the kinetic behaviour of CEES degradation mediated by KBDO has not been previously reported, making this analysis an important contribution toward understanding the reactivity of oxime-based reagents toward sulfur-containing pollutants. The kinetic evaluation was carried out by monitoring the decrease in the chromatographic peak area of CEES over time using GC–MS analysis. Since the peak area obtained from GC–MS is directly proportional to the concentration of the analyte, the degradation process was evaluated using the ratio A_0/A_t , where A_0 represents the initial peak area and A_t corresponds to the peak area at reaction time t . As shown in figure (6.d), a linear relationship is obtained between $\ln(A_0/A_t)$ and reaction time, indicating that the degradation of CEES follows pseudo-first-order kinetics under the investigated conditions. The linear regression equation obtained from the experimental data is shown in Eq. (7):

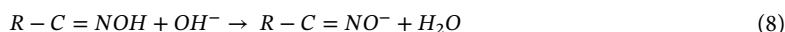
$$Y = 0.495X + 0.133 \quad (7)$$

with a correlation coefficient of: $R^2 = 0.973$. The relatively high R^2 value confirms the good agreement between the experimental data and the pseudo-first-order kinetic model.

The apparent rate constant k was determined from the slope of the linear plot and was found to be: $k = 0.495 \text{ min}^{-1}$. The half-life of CEES degradation ($t_{1/2}$) was calculated to be 1.4 min. The relatively short half-life indicates that the oximate reagent promotes a rapid degradation of CEES, highlighting the strong nucleophilic reactivity of the oximate species toward the electrophilic chloroethyl group of the CEES molecule. Overall, the kinetic analysis demonstrates that KBDO is highly effective in promoting the degradation of CEES, with rapid reaction kinetics and a short half-life, which further supports the proposed degradation mechanism discussed later. A comparison between the present system and previously reported degradation approaches is summarized in Table 5. The KBDO system exhibits a significantly shorter half-life under mild reaction conditions (room temperature, atmospheric pressure, aqueous medium, and pH 10) compared with previously reported oxidation-based methods.

Proposed degradation pathway and reaction mechanism

The proposed degradation pathway and reaction mechanism of CEES in the presence of KBDO was illustrated in figure (7). Unlike previously reported oxime-based detoxification systems that were mainly designed for military decontamination of sulfur mustard agents, the present work demonstrates that oxime chemistry can be successfully translated into environmental remediation by combining rapid degradation kinetics with practical wastewater treatment and biological assessment. This broader application highlights the versatility of nucleophilic oxime chemistry beyond conventional chemical defense technologies. The degradation of CEES can be rationalized through a nucleophilic substitution mechanism initiated by the oximate anion. The efficiency of the degradation process observed in the GC–MS experiments, particularly under alkaline conditions, indicates that the oxime functionality plays a key role in activating the degradation pathway. Under alkaline conditions, the oxime group undergoes deprotonation, generating the reactive oximate anion, which acts as a strong nucleophile as shown in Eq. (8):



Method	Pollutant	System	k (min ⁻¹)	t _{0.5} (min)	Reference
Plasma-biological process	Dimethyl sulfide	Biological treatment	-	-	46
Oxidation	DMDS/DEDS	NaClO/ClO ₂	10 ⁴ –10 ⁶	-	16
Catalytic oxydation	Thiophene	nanocatalyst	-	-	47
Photocatalytic oxidation	Polychlorinated diphenyl sulfides	Mof photocatalyst	0.175	4.4	48
Oximate system (this work)	2-Chloroethyl ethyl sulfide	KBDO	0.495	1.4	This work

Table 5. Benchmark comparison of degradation systems for sulfur-containing pollutants.

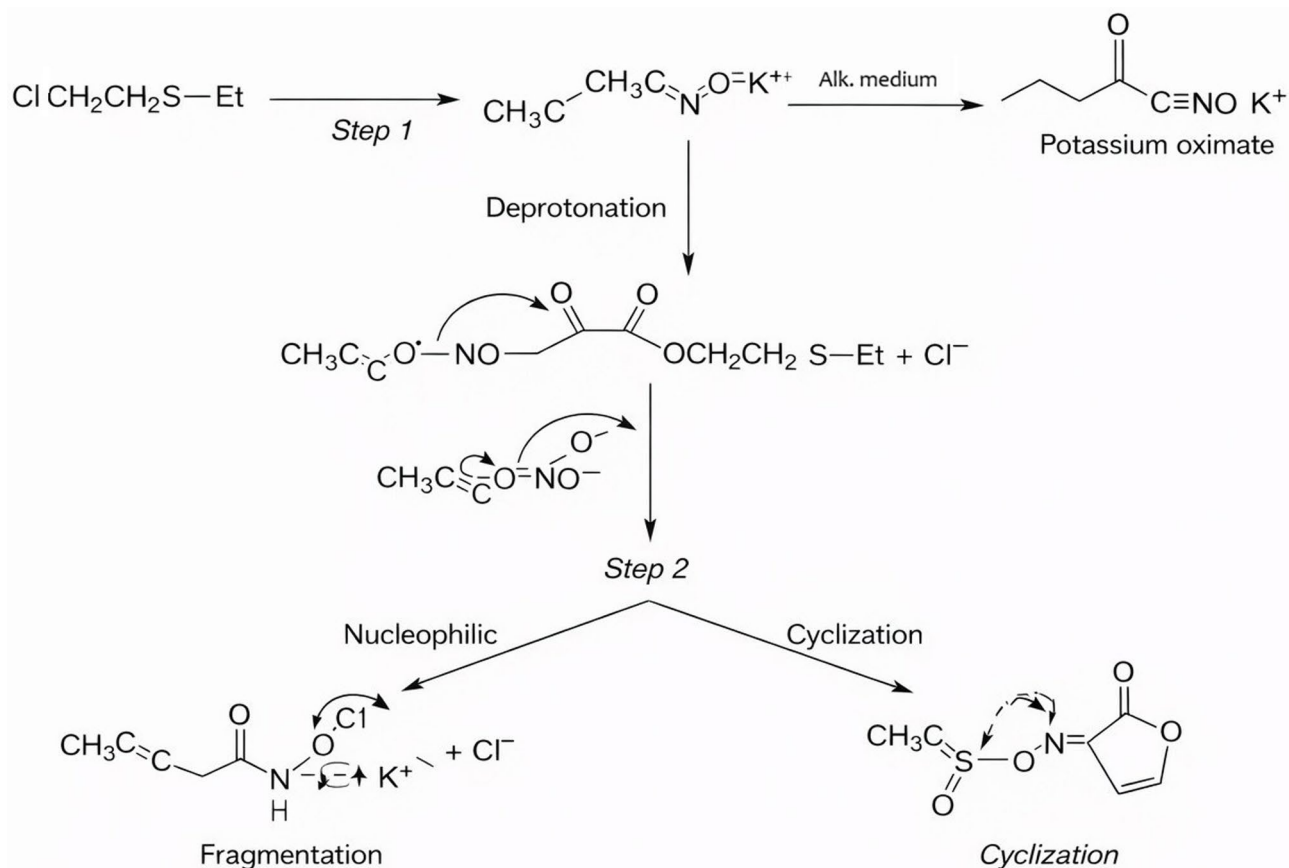
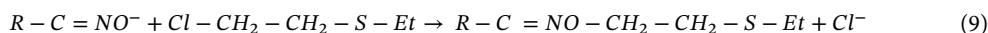


Fig. 7. Proposed degradation mechanism of CEES in the presence of potassium oximate (KBDO).

The resulting oximate species possesses a high electron density localized on the oxygen atom, which enhances its nucleophilic character. This nucleophile readily attacks the electrophilic carbon atom in the β -chloroethyl group of CEES. The presence of the electron-withdrawing chlorine atom increases the susceptibility of this carbon center to nucleophilic attack, facilitating an $\text{S}_\text{N}2$ substitution reaction as shown in Eq. (9):



This step leads to the formation of an intermediate oxime-substituted thioether species. Due to the relatively weak C-S bond and the electron-rich sulfur atom, the intermediate can undergo further transformation through bond cleavage and rearrangement reactions. Subsequent fragmentation of the thioether linkage results in the formation of smaller sulfur-containing molecules, including Diethyl sulfide, which was identified in the GC-MS chromatograms. The detection of this compound confirms the cleavage of the sulfur-containing moiety during the degradation process. In parallel, reaction intermediates containing both sulfur and nitrogen functionalities may undergo intramolecular cyclization reactions. The interaction between sulfur atoms and nitrogen-containing intermediates can generate heterocyclic products such as 1,3-Thiazolidine-4-carboxylic acid. The formation of such heterocyclic compounds suggests that nucleophilic addition and subsequent cyclization processes occur during the later stages of the reaction. The influence of pH observed in the degradation experiments further supports this mechanism. The enhanced degradation efficiency at higher pH values can be attributed to the increased formation of the oximate anion, which significantly improves the nucleophilic reactivity of the reagent. Consequently, the rate of nucleophilic substitution on the chloroethyl group of CEES increases, leading to faster degradation and greater formation of transformation products. Furthermore, the kinetic analysis presented in the previous section indicates that the degradation process follows a pseudo-first-order kinetic model, suggesting that the concentration of the reactive oximate species remains relatively constant during the reaction. This kinetic behaviour is consistent with a nucleophilic substitution mechanism in which the degradation rate depends primarily on the concentration of CEES. Finally, the combined GC-MS identification of degradation products, the observed pH dependence, and the kinetic analysis collectively support a degradation pathway involving oximate-mediated nucleophilic substitution followed by fragmentation and cyclization reactions. The proposed mechanism provides a comprehensive explanation for the observed transformation of CEES into sulfur-containing intermediate and heterocyclic products. During the degradation process, KBDO functions as

a nucleophilic reagent and is expected to be progressively consumed through its direct reaction with CEES rather than acting as a catalytic species. Accordingly, the residual concentration of KBDO is anticipated to decrease substantially as the reaction proceeds. Although the residual oximate concentration was not experimentally quantified in the present study, the nearly complete disappearance of CEES together with the formation of degradation products observed by GC–MS strongly suggests extensive consumption of the active oximate species during the reaction. Consequently, the antimicrobial activity reported herein should not be exclusively attributed to residual KBDO, as both the remaining reagent, if present, and the generated degradation products may collectively contribute to the observed biological response. Quantitative determination of residual KBDO after treatment and its independent biological assessment will be the subject of future investigations to further clarify its contribution.

In vitro Antimicrobial Activity of KBDO against marine microbial pathogens

The antimicrobial activity of KBDO was quantitatively evaluated based on optical density (OD₆₀₀) measurements, which provide a direct indication of microbial growth. The inhibition efficiency was calculated by comparing treated samples with untreated controls, following standard broth microdilution protocols^{30,31}. As summarized in Table 6, KBDO exhibited a clear concentration-dependent antimicrobial effect across all tested microorganisms. A significant reduction in OD values was observed at a concentration of 1.0 mg mL⁻¹, indicating effective suppression of microbial growth. Among the tested strains, *P. aeruginosa* showed the highest sensitivity, where the OD decreased from 0.263 to 0.024, corresponding to an inhibition efficiency of approximately 90.9%. Similarly, *P. fluorescens* and (*A. hydrophila*) exhibited inhibition efficiencies of 52.1% and 47.0%, respectively. Moderate inhibition was observed for *E. faecalis* (51.0%) and *V. cholerae* (40.4%), while relatively lower inhibition values were recorded for *S. aureus* (34.8%), *E. coli* (32.7%), and (*B. subtilis*) (31.4%). The yeast (*C. albicans*) also showed measurable antifungal activity with an inhibition efficiency of 26.7%.

However, while certain strains like *P. aeruginosa* exhibited high sensitivity, concluding that KBDO universally penetrates and destabilizes outer lipopolysaccharide membranes to facilitate intracellular interactions would be overly speculative. In particular, *E. coli* a classic gram-ve bacillus featuring an outer membrane displayed a comparatively low inhibitory response (32.7%), indicating that susceptibility is not strictly dictated by Gram classification or simple outer membrane presence, and that other resistance mechanisms or intrinsic factors modulate the antimicrobial efficacy⁴⁹.

The observed antimicrobial behavior can be directly correlated with the chemical reactivity of the oximate system. As discussed in the degradation mechanism section, KBDO exists predominantly in its active nucleophilic form (oximate anion) under alkaline conditions, which enhances its ability to interact with both electrophilic pollutants and biological macromolecules. The oximate species, characterized by high electron density and strong nucleophilicity, may induce membrane disruption or interfere with essential enzymatic systems within microbial cells, ultimately leading to growth inhibition^{50,51}.

While these traditional explanations regarding membrane disruption and enzymatic interference remain largely speculative, the high nucleophilicity of KBDO and its reaction products strongly points toward specific potential molecular targets. These may include thiol-containing enzymes (which are highly vulnerable to nucleophilic attack), essential cellular nucleophiles, or DNA-related targets that disrupt genetic integrity and replication. Although conducting empirical validations such as membrane permeability assays utilizing propidium iodide uptake or targeted enzymatic inhibition studies is beyond the current scope of this work, acknowledging and exploring these precise molecular interaction sites provides a critical framework for understanding the oximate-driven antimicrobial process⁵².

Furthermore, the higher susceptibility observed for several Gram-negative bacteria suggests that KBDO can effectively penetrate or destabilize outer membrane structures, thereby facilitating intracellular interactions. Similar behavior has been reported for chemically reactive antimicrobial agents capable of inducing cellular stress through membrane permeabilization and metabolic disruption^{53,54}. In addition, nucleophilic or oxidative stress generated during the reaction may contribute to enzyme inactivation and loss of cellular integrity, further enhancing antimicrobial performance⁵¹. Overall, these findings demonstrate that the antimicrobial activity of KBDO is intrinsically linked to its chemical reactivity, highlighting its dual functionality as both a degradation agent for sulfur-containing pollutants and an effective antimicrobial compound. This multifunctional behavior is particularly advantageous for wastewater treatment applications, where simultaneous removal of chemical contaminants and microbial load is required.

Application of KBDO for treatment of real industrial wastewater

To investigate the applicability of the oximate system under realistic environmental conditions, the degradation behaviour of CEES was examined in a real industrial wastewater sample collected from a petroleum processing facility. Since the optimized degradation efficiency was achieved at pH 10, the refinery wastewater sample was adjusted from its initial pH of 7.3 to pH 10 prior to KBDO addition. This standardization ensured that the degradation performance observed in the real matrix reflected the intrinsic influence of wastewater constituents rather than variations in solution alkalinity. The chromatograms of the untreated and treated samples are presented in figure 8, where the chromatographic signal corresponding to CEES and the peaks associated with its degradation products are highlighted. The chromatogram of the untreated wastewater shows a well-defined peak corresponding to CEES at a retention time of approximately 6 min, confirming the presence of the target sulfur-containing compound within the complex wastewater matrix. The relatively high intensity of this peak reflects the initial concentration of the contaminant prior to treatment. Following the addition of KBDO and incubation under the same reaction conditions used in the laboratory degradation experiments, a pronounced decrease in the CEES signal was observed. Quantitative analysis based on the integrated chromatographic peak areas revealed that the intensity of the CEES peak decreased by approximately 97%, indicating highly efficient

Marine pathogens	Type	0.03 mg/mL	0.07 mg/mL	0.1 mg/mL	0.2 mg/mL	0.3 mg/mL	0.4 mg/mL	0.5 mg/mL	1.0 mg/mL	Control (-)	Control (+)	Inhibition (%)
<i>S. aureus</i>	Gram +ve	0.490	0.450	0.445	0.443	0.440	0.436	0.435	0.352	0.540	0.016	34.8
<i>E. coli</i>	Gram -ve	0.398	0.366	0.357	0.352	0.350	0.344	0.342	0.298	0.443	0.016	32.7
<i>P. aeruginosa</i>	Gram -ve	0.278	0.281	0.279	0.274	0.195	0.183	0.174	0.024	0.263	0.021	90.9
<i>B. subtilis</i>	Gram +ve	0.291	0.301	0.283	0.279	0.274	0.269	0.263	0.194	0.283	0.014	31.4
<i>E. faecalis</i>	Gram +ve	0.323	0.293	0.248	0.246	0.241	0.239	0.235	0.201	0.410	0.008	51.0
<i>K. pneumoniae</i>	Gram -ve	0.497	0.455	0.451	0.441	0.433	0.416	0.410	0.374	0.453	0.013	17.4
<i>V. cholerae</i>	Gram -ve	0.423	0.307	0.320	0.277	0.214	0.212	0.205	0.193	0.324	0.009	40.4
<i>C. albicans</i>	Yeast	0.456	0.456	0.452	0.437	0.396	0.388	0.377	0.286	0.390	0.010	26.7
<i>P. fluorescens</i>	Gram -ve	0.141	0.135	0.120	0.120	0.118	0.117	0.112	0.093	0.194	0.004	52.1
<i>S. agalactiae</i>	Gram +ve	0.256	0.253	0.248	0.246	0.241	0.238	0.218	0.178	0.261	0.002	31.8
<i>A. hydrophila</i>	Gram -ve	0.171	0.168	0.151	0.136	0.130	0.129	0.109	0.096	0.181	0.001	47.0

Table 6. Concentration-dependent antimicrobial activity of KBDO against selected microorganisms based on optical density (OD₆₀₀) measurements. Inhibition (%) was calculated relative to the negative control at the highest tested concentration (1.0 mg mL⁻¹).

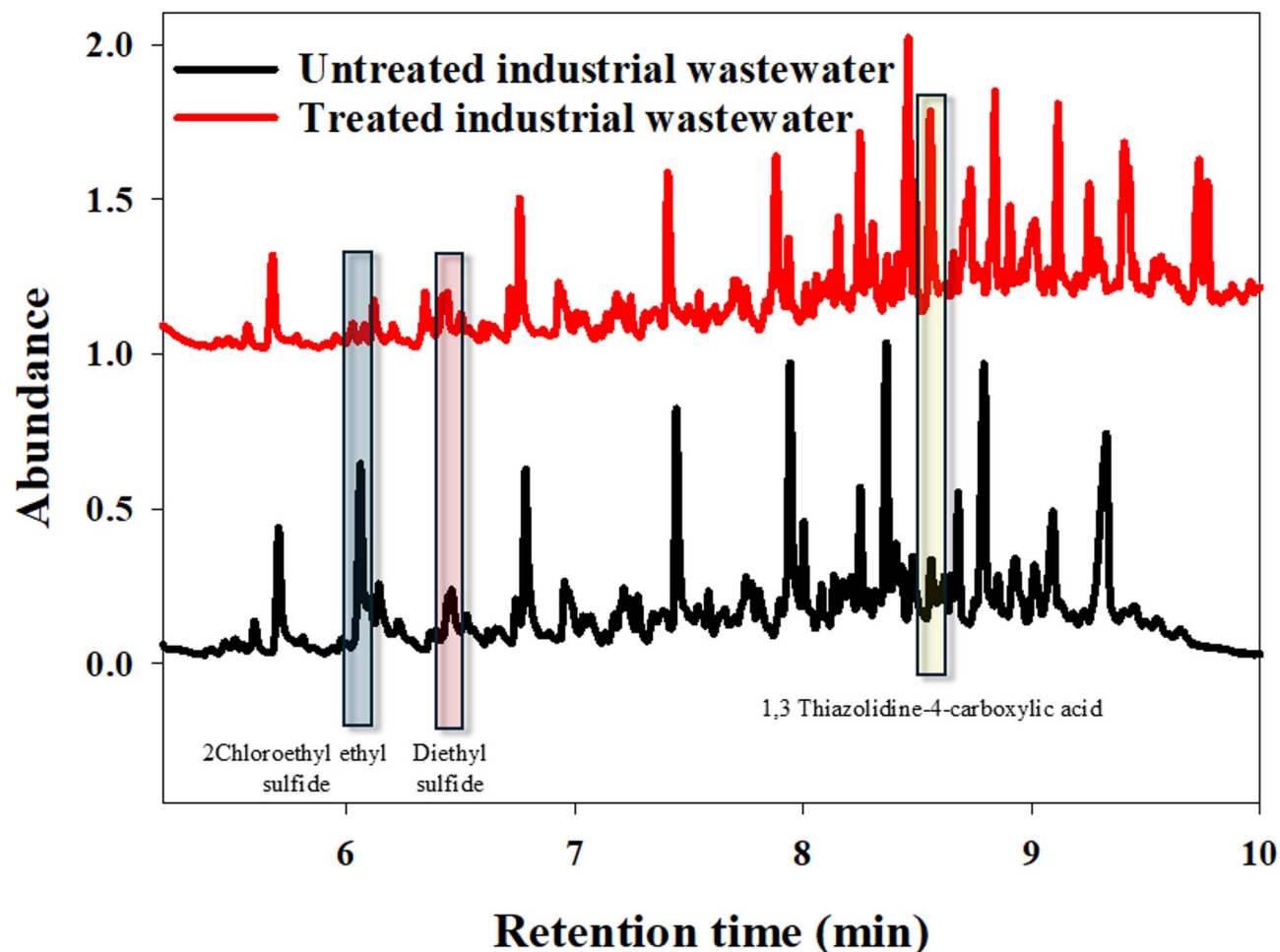


Fig. 8. GC-MS chromatograms of real petroleum refinery wastewater samples before and after treatment with potassium oximate (KBDO).

degradation of the parent compound even in the presence of competing organic constituents present in the wastewater matrix. Simultaneously, several new peaks appeared in the chromatogram of the treated sample. These peaks correspond to the degradation products previously detected in the controlled laboratory experiments. The emergence of these signals together with the disappearance of the parent CEES peak confirms that the oximate reagent promotes a parent-to-product transformation rather than simple suppression of the chromatographic signal. The degradation trend observed in the wastewater matrix is consistent with the kinetic behaviour determined in the laboratory experiments, where CEES degradation followed pseudo-first-order kinetics with a rate constant of $k \approx 0.495 \text{ min}^{-1}$ and a half-life of approximately 1.4 min. The preservation of this reactivity in real wastewater demonstrates that the oximate reagent retains its nucleophilic activity toward sulfur-containing electrophilic compounds even in chemically complex aqueous environments. Based on the identified degradation pathway and the disappearance of the parent CEES molecule, the degradation products are expected to exhibit a lower toxicological risk than the parent compound. However, comprehensive ecotoxicological assessment using standardized bioassays is required to experimentally confirm the detoxification efficiency of the treated solution. The high reactivity and biological toxicity of CEES primarily arise from the presence of the chloroethyl group, which can generate highly reactive sulfonium intermediates capable of alkylating biomolecules. During the oximate-induced transformation, this reactive functionality is either oxidized or cleaved, leading to the formation of more stable derivatives such as oxidized sulfur species and hydrolysed fragments. These transformation products generally possess reduced electrophilicity and therefore lower biological reactivity, suggesting that the degradation process not only removes the parent pollutant but also contributes to partial detoxification of the contaminated water matrix. Petroleum refinery wastewater contains a complex mixture of organic constituents, including sulfur-containing compounds, aromatic hydrocarbons, oxygenated species, and other electrophilic contaminants that may potentially compete with CEES for reaction with KBDO. As a nucleophilic reagent, KBDO is expected to preferentially react with highly electrophilic substrates such as CEES owing to the strong leaving ability of the β -chloroethyl group, which facilitates rapid nucleophilic substitution. Nevertheless, the presence of additional reactive electrophilic species in real wastewater may result in partial reagent consumption through competing side reactions, potentially reducing the effective availability of KBDO toward the target contaminant. Despite this possibility, the high CEES removal efficiency achieved

in the petroleum refinery wastewater demonstrates that matrix interference did not significantly compromise the degradation performance under the investigated conditions. Future studies involving individual refinery wastewater constituents and competitive reaction experiments will provide further insight into the reaction selectivity of KBDO in complex environmental matrices. Overall, the results demonstrate that KBDO can efficiently degrade CEES in real industrial wastewater, achieving 97% pollutant removal, thereby confirming the feasibility of extending the oximate-based detoxification strategy beyond controlled laboratory systems to more realistic environmental conditions. The oximate reagent is largely consumed during the degradation process, and any remaining traces can be readily removed using conventional polishing treatments such as activated carbon adsorption or mild oxidation (room temperature, atmospheric pressure, aqueous medium, and pH 10), which are routinely applied in water treatment systems.

The antimicrobial performance of KBDO was further evaluated using a real petroleum refinery wastewater sample in order to assess its effectiveness under environmentally relevant conditions. The results are illustrated qualitatively in figure (9) and quantitatively in Table (7). As shown in figure (9), the untreated wastewater samples exhibited dense and heterogeneous microbial growth across different selective and differential media, indicating the presence of diverse microbial populations. In contrast, the treated samples showed a pronounced reduction in colony density, with several plates exhibiting nearly complete suppression of visible growth. This clear visual distinction confirms the strong antimicrobial activity of KBDO in complex wastewater matrices. To further support these observations, microbial populations were assessed using standard culture-based methods, and the results are summarized in Table (7). A significant reduction in total bacterial count was observed after treatment, as indicated by reduced growth on nutrient agar. Similarly, key pathogenic indicators such as *E. coli*, *Staphylococcus sp.*, *Pseudomonas sp.*, and *Vibrio sp.* exhibited clear reductions in growth following treatment, confirming the broad-spectrum antimicrobial activity of the oximate system. In contrast, fungi, *Aeromonas sp.*, total coliforms, *Enterococcus sp.*, *Salmonella sp.*, and *Shigella sp.* were not detected in both untreated and treated samples, suggesting that these microorganisms were either absent or present below the detection limit. The observed microbial reduction is consistent with the chemical reactivity of the oximate system discussed earlier. Under alkaline conditions, KBDO exists predominantly in its active nucleophilic form (oximate anion), which enables interaction with both electrophilic organic pollutants and biological macromolecules within microbial cells. This highly reactive species can disrupt cell membranes, alter permeability, and interfere with intracellular enzymatic systems, ultimately leading to microbial inactivation. The pronounced reduction observed for Gram -ve bacteria such as *Pseudomonas sp.* and *E. coli* suggests that the oximate species can effectively penetrate or destabilize outer membrane structures, enhancing its antimicrobial efficiency. In addition, the rapid degradation of organic pollutants observed in laboratory experiments contributes to modifying the chemical environment of the wastewater, thereby limiting microbial proliferation. This indicates a synergistic effect, where chemical degradation and antimicrobial activity occur simultaneously. Similar trends have been reported in wastewater treatment processes, where chemical treatments significantly enhance microbial reduction efficiency^{55,56}. The strong agreement between visual observations (Fig. 9), quantitative microbial reduction (Table 7), and previously discussed degradation kinetics confirms that KBDO operates as a multifunctional treatment system capable of simultaneous pollutant removal and microbial inactivation under realistic conditions.

Limitations of study

- One limitation of the present work is that the residual concentration of KBDO after completion of the degradation reaction was not directly quantified. Therefore, although the reaction mechanism indicates extensive reagent consumption, the exact contribution of residual KBDO to the observed antimicrobial activity cannot be completely distinguished from that of the degradation products. Future work will focus on quantitative residual oximate analysis using appropriate chromatographic methods together with independent toxicity and antimicrobial evaluation.
- Another limitation of this study is that environmental detoxification was inferred from chemical degradation and product identification rather than direct ecotoxicological testing. Although GC-MS confirmed the complete transformation of CEES into less reactive products, dedicated toxicity assays will be performed in future work to experimentally verify the environmental safety of the treated solution.
- Although the proposed KBDO system demonstrated rapid and efficient CEES degradation, several limitations should be acknowledged. The residual concentration of KBDO after treatment was not quantitatively

Microorganism/Medium	Pre-treatment	Post-treatment	Observation
Nutrient agar (Total count)	High growth	Reduced	Significant reduction
<i>E. coli</i>	Detected	Reduced	Significant reduction
<i>Staphylococcus sp.</i>	Detected	Reduced	Significant reduction
<i>Pseudomonas sp.</i>	Detected	Reduced	Significant reduction
<i>Vibrio sp.</i>	Detected	Reduced	Significant reduction
Fungi/ <i>Aeromonas sp.</i>	Not detected	Not detected	No detection
Total coliform/ <i>Enterococcus sp.</i>	Not detected	Not detected	No detection
<i>Salmonella/Shigella</i>	Not detected	Not detected	No detection

Table 7. Comparative microbial counts in real wastewater before and after treatment.

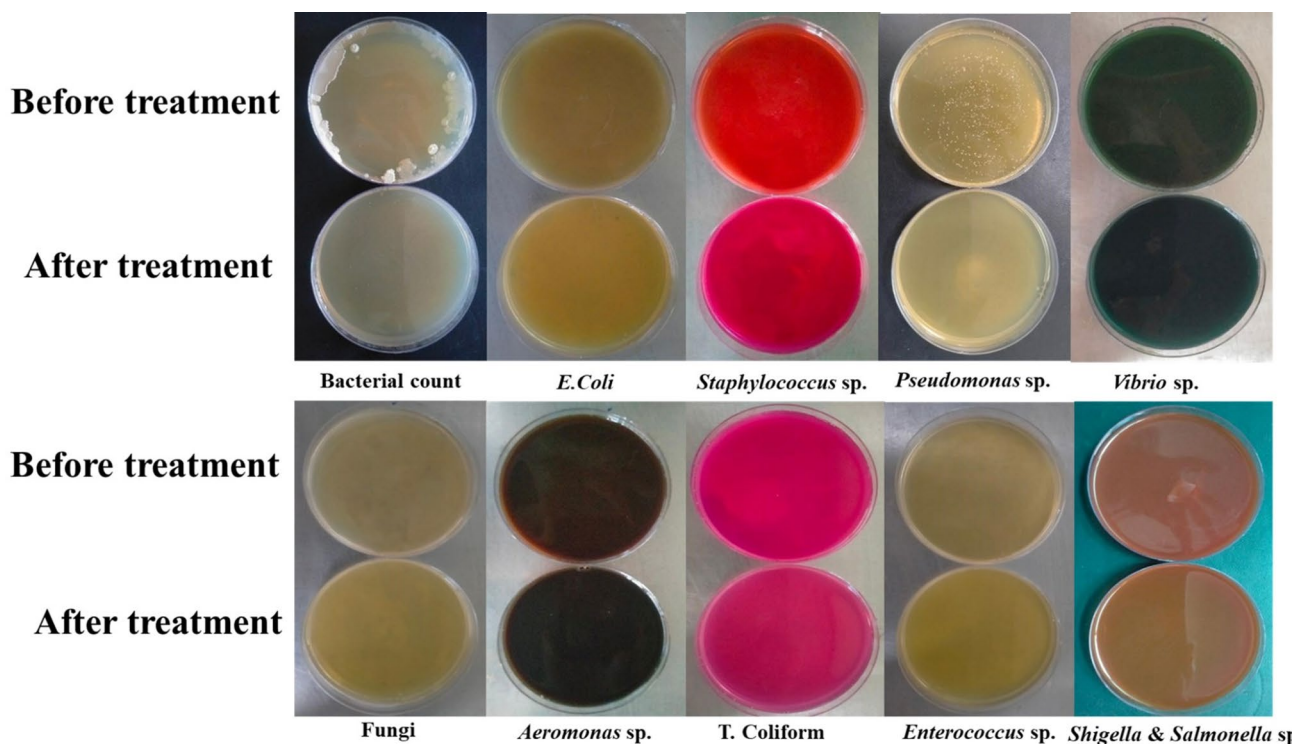


Fig. 9. Microbial growth on selective media before and after treatment with potassium oximate (KBDO).

determined, and environmental detoxification was inferred from chemical degradation rather than direct ecotoxicological assays. In addition, the selectivity of KBDO toward CEES in the presence of other reactive wastewater constituents was not systematically investigated. Future studies should therefore focus on residual reagent analysis, standardized toxicity evaluation, competitive reaction studies, and pilot-scale techno-economic assessment.

- d) Furthermore, the stoichiometric relationship between KBDO and CEES was not systematically optimized, and the influence of KBDO concentrations exceeding 20% remains to be investigated.
- e) In addition, the long-term stability of KBDO in alkaline aqueous solution during prolonged storage was not evaluated and should be investigated in future studies.
- f) In addition, the long-term stability and environmental fate of the identified degradation products were not monitored and should be investigated in future studies.

Practical scalability and economic considerations

The proposed KBDO-based degradation system offers several practical advantages for scale-up applications. Unlike many conventional oxidation processes, it operates under mild aqueous conditions (room temperature, atmospheric pressure, aqueous medium, and pH 10) without external energy input or transition-metal catalysts, while achieving complete CEES degradation within approximately 10 min. These features may reduce operational complexity and energy demand. Since KBDO functions as a reactive nucleophilic reagent rather than a catalyst, it is expected to be consumed during the degradation process, making reagent consumption an important economic consideration. Although a detailed techno-economic assessment and reagent regeneration were beyond the scope of this study, the rapid reaction kinetics, simple operating conditions, and absence of sludge generation support the practical potential of the proposed process. Future studies should focus on reagent utilization efficiency, regeneration strategies, and pilot-scale techno-economic evaluation to further improve process sustainability.

Conclusion

In this study, a rapid and efficient strategy for the degradation of sulfur-containing pollutants in aqueous systems has been developed using KBDO. Using 2-Chloroethyl ethyl sulfide (CEES) as a representative thioether model, the proposed system demonstrated fast and effective pollutant transformation under mild conditions (room temperature, atmospheric pressure, aqueous medium, and pH 10). The degradation process followed pseudo-first-order kinetics with a high-rate constant ($k = 0.495 \text{ min}^{-1}$) and a short half-life ($t_{0.5} = 1.4 \text{ min}$), highlighting the exceptional reactivity of the oximate species. Mechanistic investigations confirmed that the degradation proceeds via a nucleophilic substitution pathway initiated by the oximate anion, followed by fragmentation and formation of secondary products, as supported by GC-MS and spectroscopic analyses. Importantly, the system maintained high performance in a real petroleum refinery wastewater matrix, achieving approximately 97% removal of the target pollutant, demonstrating its robustness under complex environmental conditions. Beyond

chemical degradation, the oximate system exhibited notable antimicrobial activity against a range of microbial strains, indicating its dual functionality for simultaneous chemical and biological remediation. Compared to conventional treatment approaches that rely on strong oxidants, catalysts, or external energy input, the present system offers a simple, fast, and environmentally benign alternative operating under ambient conditions. Although competitive reactions with other electrophilic constituents may occur in complex wastewater matrices, the proposed KBDO system maintained high degradation efficiency under real refinery wastewater conditions, demonstrating promising practical robustness. Finally, the proposed process shows promising potential for scale-up; however, comprehensive techno-economic analysis together with process optimization at pilot scale will be required before full-scale implementation.

Data availability

All data is available by the corresponding author: hussein.mohamed4544@yahoo.com.

Received: 4 July 2026; Accepted: 26 August 2026

Published online: 17 September 2026

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Hussein Oraby & Ghada E. Hegazy: Conceptualization, Methodology, Writing, Editing and reviewing the final copy. Ahmed saeed Abo elfath & Mohamed Atef Abdelfatah: Methodology, Investigation, Data collection and analysis.

Funding

Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).

Declarations

Competing interests

The authors declare no competing interests.

Consent for publication

All authors approved for the publication.

Additional information

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